

Data Science/Analysis with Python Internship– Task 1

Student Performance dataset

Task Description

Your first step into **Data Science**! In this task, you'll explore the **Student Performance dataset**  and use Python to **clean, analyze, and visualize** it.

This will teach you the **workflow every data scientist follows**: load → clean → analyze → visualize → conclude.

Responsibilities (What to Build)

1. Load Dataset

- Use pandas to read student-mat.csv dataset ([Download here](#)).

2. Explore & Clean Data

- Check missing values (.isnull()).
- Remove duplicates.
- Inspect dataset shape & dtypes.

3. Analysis Questions

- Average final grade (G3).
- How many students scored above 15?
- Is study time correlated with performance?
- Which gender performs better on average?

4. Visualizations

- Histogram of grades.
- Scatterplot: study time vs grades.
- Bar chart: male vs female average score.

5. Documentation

- Use Markdown in Jupyter Notebook to explain each step.

Skills Gained

- ✓ Data loading & cleaning (Pandas)
 - ✓ Statistical analysis with NumPy
 - ✓ Data visualization with Matplotlib/Seaborn
 - ✓ Correlation analysis
 - ✓ Writing reports with Jupyter Notebook
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Sample Starter Code

python

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("student-mat.csv")

# Analysis
print("Average Final Grade:", df['G3'].mean())

# Visualization
plt.hist(df['G3'], bins=10, color='skyblue', edgecolor='black')
plt.title("Distribution of Final Grades")
plt.xlabel("Grade")
plt.ylabel("Frequency")
plt.show()
```

Resources

Free Video Tutorials

- [Pandas Crash Course – Corey Schafer](#)
- [Data Science Crash Course – FreeCodeCamp](#)

Reference Docs

- [Pandas Official Docs](#)
- [Matplotlib Docs](#)
- [Seaborn Docs](#)

Tools

- [Google Colab](#) → Free cloud-based Jupyter Notebook
- [Kaggle](#) → Datasets + practice environment

Deliverable

A Jupyter Notebook `student_analysis.ipynb` with:

- Clean dataset
- Answers to analysis questions
- Visualizations
- Markdown explanations